

PARKER HUNT

EDUCATION

2021-present

Boston College

Morrissey College of Arts and Sciences — Brighton, Massachusetts

Ph.D. candidate in physics doing atmospheric research.

2016-2020

The Ohio State University

College of Arts and Sciences — Columbus, Ohio, United States of America

B.S. Physics 2020- 3.85(Magna Cum Laude) with research distinction in physics

Physics Major GPA 3.9 out of 4

Math Minor

2016 High School Diploma

Hilliard Davidson — Hilliard, Ohio, United States of America

Graduated with a 4.3 (AP classes graded on 5.0)

PROFESSIONAL EXPERIENCE

08/2022 to present

Research with Yi Ming (Atmospheric Physics)

Boston College- Brighton, Massachusetts

Investigating isolating well known climate phenomenon with idealized mathematical models. Currently looking into East African Waves and their ability to become hurricanes that hit the east coast of the United States. Also investigating the effects of CO₂ on Hadley and Walker circulation. Work involves running idealized climate models on the BC super computer which runs on fortran backend and interacted with by a python frontend. All worked with on the unix server of BC's supercomputer. Minimal assistance from other students however have mentored an undergraduate specifically at the computing elements.

08/2021 to present

Teaching Assistant

Boston College – Brighton Massachusetts

Taught and assisted teaching of the six courses some multiple times listed below: Intro physics lab electro magnetism and mechanics, classical mechanics for engineers, electro magnetism for medical students, and physics of the environment. Communicated with students with varying backgrounds to try and educate physics ideas as well as some environmental concerns. Graded all homeworks and provided feedback

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04/2019 to 2021

Research with Ciriya Jayaprakash (Nonlinear-Dynamics in Ecology)

Ohio State University- Columbus, Ohio

Studied spatially extended, stochastic, dynamical models of ecological systems (semi-arid regions) to understand the motion of interfaces between different states and the stability of a given state with respect to domains of a competing state. These are non-equilibrium systems and the study primarily used numerical simulations of discretized models in two dimension and continuum (pde's) in one dimension. All simulated in c++ and data analyzed in Mathematica. Also analyzed weather data collected in Columbus using python dating back to 1900. We were specifically interested in looking for trends known to be associated with global climate change on a smaller scale. Work was independently paced with minimal assistance from graduate students.

08/2020 to 2021

Teaching Assistant for Introductory Calculus Based Mechanics

Ohio state University- Columbus, Ohio

Ran freshman level recitations and labs where students would complete group work and take quizzes. Mediated and facilitated group learning. Worked one on one with students over zoom to create valuable distance learning experiences. Graded assignments ranging from labs to show work homework and exams about 100 assignments per week.

08/2018 to 05/2020

Teaching Assistant to Professor Gregory Kilcup for 2300-2301 Intermediate Mechanics
Ohio State University- Columbus

Facilitated discussion in a flipped classroom setting with 200 students at a sophomore level mechanics course based on Morin's mechanics textbook. Graded 200+ homework assignments per week. Worked in small groups and individually with students tutoring them in the material. Ran midterm review sessions with 40+ students covering topics covered on the midterms.

04/2017 to 2019

Undergraduate research with Andrew Heckler (Physics Education Research)

Ohio State University- Columbus

Created hundreds of questions designed to test students' essential skills to lessen cognitive overload when attempting physics problems which are currently being implemented in the intermediate mechanics course at OSU. Met with Dr Heckler and Dr Kilcup to organize ideas so we could best implement this in Dr Kilcup's course. Analyzed data on a different project about students' estimates for physical values such as the weight of a car in kilograms or the size of football field in meters, specifically looking for correlations in accuracy to known values and grades earned in courses. Work was independently paced with minimal assistance from graduate students.

06/2017 to 11/2018

Undergrad research as part of Jay Gupta's group (Condensed Matter Experiment)

Ohio State University- Columbus, Ohio

Set up a tabletop Raman spectrometer to image graphene and other semi-conductors. Participated in weekly discussions with graduate and postdoc group members about current research in the field and the research they were doing.

06/2017 to 2/2018

Student worker at the Physics Computing Facility

Ohio State University- Columbus

Handled IT work in the Physics Research Building and Smith Lab. Solved computer problems encountered by faculty and students. Made sure all Ohio State University assets are safe and up to date when needed.

03/2017 to 11/2018 Undergrad participant in the Graphene Factory
Ohio State University – Columbus, Ohio
Handled delicate 2-D materials to evaluate quality. Used Zeiss Optical Microscope to get accurate images of mono layered graphene. Worked to evaluate properties of graphene made from different methods.

05/2015 to 08/2017 Intern in Education Manufacturing
Buckeye Facilitation — Hilliard, OH
As a high school student, helped the business with whatever they needed help with. assembled lab kits and supplies for schools. Occasionally delivered the supplies to schools when needed.

EXTRA COURSEWORK

Physics Education Research - Physics department graduate level

Applied Differential Geometry in General Relativity- Math department graduate level

Big Data Analytics and AI methods- Physics department graduate level

Computational Physics- Physics department graduate level

Geophysical modeling- Earth Science department graduate at BC

SKILLS

CAD modeling with Autodesk

Familiar with coding in C++, Python, Mathematica, Java, fortran

ACCOMPLISHMENTS

James Smith Scholarship 2019

Summer scholarship for research 2018

Freshman physics award for academic success

Dean's List OSU 7 semesters (Out of 8)

Plaque granted by the Hilliard city council for helping to clean up ponds in the area